Three explicit unitary examples in rank four

A certificate for the examples in Section 7 of Leemans–Vandenschrick

The source reports three pairs of enantiomorphic chiral polytopes for $\mathrm{PSU}_3(17)$, of types $[3,7,3]$, $[3,8,3]$ and $[3,9,3]$. It gives neither matrices nor a search certificate for this sentence. We supply explicit matrices and verify these three positive examples. We do not infer that other types are absent, and we do not treat the source's historical question for larger fields as currently open.

Let $F = \mathbb{F}_{17}[\beta]$, where $\beta^2 = 3$. Since 3 is a nonsquare modulo 17, the nontrivial field automorphism is $\overline{\beta} = -\beta$. We use the standard Hermitian form $z^*z = \overline{z}^t z$. For a nonisotropic column z, put

$$R_z = -I + \frac{2zz^*}{z^*z}.$$

This matrix fixes z, negates its orthogonal complement, and has unitary determinant one. Put

$$u = R_{(4\beta,0,1)^t}, \qquad w = R_{(0,0,1)^t}.$$

For the three choices below, set $v = R_{(x,y,1)^t}$ and

$$a = wv, \qquad b = vwu, \qquad c = uw. \tag{1}$$

$\lvert b\rvert$	x	y	$\mathrm{Tr}_{F/\mathbb{F}_{17}}(x/(4\beta))$
7	$3+4\beta$	$1+3\beta$	2
8	$7+\beta$	$1+3\beta$	9
9	$8+6\beta$	$4+\beta$	3

(2)

Here and below orders are projective orders when the matrices represent unitary projective transformations.

Theorem 1. *For each row of (2), the images of (a,b,c) generate $\mathrm{PSU}_3(17)$ and satisfy the full string rotation intersection property. They define a rank-four chiral polytope of the indicated type $[3,|b|,3]$. The three types give three distinct pairs of enantiomorphic markings.*

The proof combines exact, small-input finite group calculations with a full-automorphism argument. We retain the standard rotation correspondence for abstract polytopes, the classical order and faithful isotropic-point action of $\mathrm{PSU}_3(17)$, and its classical automorphism description

$$\mathrm{Aut}(\mathrm{PSU}_3(17)) = \mathrm{PGU}_3(17) \rtimes \langle \overline{\ } \rangle. \tag{3}$$

For this prime field the last factor has order two. The inverse-transpose automorphism on unitary matrices is the same field map, so no separate graph coset is omitted. These are attributed classical group inputs, not new classifications asserted here. GAP's exact finite permutation group algorithms supply the group orders and subgroup intersections specified below; neither the source search nor a named-group constructor supplies them.

The interval relations and actual subgroup checks

All four nontrivial centres in (2) and the definition of u have Hermitian norm four. In particular all denominators are nonzero. The first and last rotations have order three. One may check this directly, or use the formula for two unitary involutions: their product has eigenvalue one and trace $4|z^*e_3|^2/(z^*z) - 1$. Here that trace is zero, so the remaining two eigenvalues are primitive cube roots. Equation (1) gives identically

$$ab = u, \qquad bc = v, \qquad abc = w. \tag{4}$$

The other required interval products are therefore involutions. Exact multiplication gives the middle orders in (2).

The finite calculation constructs all $4,914$ isotropic projective points directly. They are the normalized vectors

$$(1, s, t) \quad \text{with } 1 + s\bar{s} + t\bar{t} = 0, \qquad (0, 1, s) \quad \text{with } 1 + s\bar{s} = 0.$$

The remaining normalized point $(0, 0, 1)$ is not isotropic. These lists contain every projective point of norm zero exactly once. The program uses row representatives and right multiplication, preserving these Hermitian equations; the abstract matrix products are unchanged. The action of each literal matrix in (1) on these lists is computed, without identifying it with a preselected permutation generator. All three generated permutation groups have order

$$\frac{17^3(17^3 + 1)(17^2 - 1)}{3} = 2,317,678,272.$$

They are subgroups of the faithful projective unitary action; equality with its classical order proves generation of the whole group. The two adjacent subgroups have respective orders

$\lvert b\rvert$	$\lvert\langle a, b\rangle\rvert$	$\lvert\langle b, c\rangle\rvert$
7	168	168
8	2448	2448
9	2448	2448

No group identification from these last orders is needed.

We check every indexed intersection, rather than infer it from these orders. For $I \subseteq \{0, 1, 2, 3\}$ let G_I be generated by those of the following six elements whose endpoint pair lies in I:

$$(01) : a,\ (02) : ab,\ (03) : abc,\ (12) : b,\ (13) : bc,\ (23) : c.$$

The verifier constructs all sixteen actual permutation subgroups and checks $G_I \cap G_J = G_{I\cap J}$ for all 256 ordered pairs. For nested index sets it verifies actual subgroup containment; for all other pairs it computes the actual intersection. Adding the trivial endpoint conventions in the source does not change any of these groups. The three complete tests pass, giving precisely the rotation intersection property.

Excluding every reversing automorphism

Lemma 2. *Let u, w be the two matrices above. If $v = R_{(x,y,1)^t}$ and $\mathrm{Tr}_{F/\mathbb{F}_{17}}(x/(4\beta)) \neq 0$, no automorphism of $\mathrm{PSU}_3(17)$ fixes u, w and sends v to wvw.*

Proof. Write $\alpha = 4\beta$. A linear matrix commuting with w preserves its centre e_3 and its axis $\langle e_1, e_2\rangle$. Commuting also with u, whose centre has nonzero first and third coordinates, forces its two off-diagonal entries in the e_1, e_2 block to vanish and its first and third diagonal entries

to agree. Thus its projective form is $D_r = \mathrm{diag}(1, r, 1)$. It is a unitary similitude exactly when $r\bar{r} = 1$.

Fixing the projective involutions really does give these commuting matrix conditions: their determinant-one involutory lifts are unique, since the scalar centre has odd order three. Conjugation by a projective unitary representative, and the field map, preserve these lifts. The semilinear map given by field conjugation followed by $D_0 = \mathrm{diag}(\alpha/\bar{\alpha}, 1, 1)$ fixes u, w. Composing with it reduces every semilinear centralizer element to the linear case. Hence the full centralizer in (3) consists of the eighteen D_r and the eighteen maps $D_r D_0^{-}$.

The centre of wvw is $(-x, -y, 1)^t$. Its last coordinate is already normalized, so equality of centres under a linear centralizer element requires $x = -x$, hence $x = 0$. Under a semilinear element it requires $(\alpha/\bar{\alpha})\bar{x} = -x$, or

$$x/\alpha + \overline{x/\alpha} = 0.$$

Both possibilities contradict the hypothesis. This accounts for every automorphism in (3), independently of the second coordinate equation. □

A reversing automorphism for (1) would send a to a^{-1} while fixing $ab = u$ and $abc = w$, by the standard chiral criterion. Since $v = wa$, it would send v to $wa^{-1} = wvw$. The three traces in (2) are nonzero. The lemma therefore excludes all reversing automorphisms. With generation, (4) and the verified indexed intersections, the rotation correspondence proves the theorem. Distinct displayed types distinguish the three enantiomorphic pairs.

Certificate and scope

The supplied GAP program builds the field, the three homologies and all isotropic points afresh. It checks group membership, all exact rotation orders, all 768 indexed intersections and the 36 possible reversers for each row. An independent elementary field implementation checks the matrix identities, types and traces without importing GAP. Two additional controls prevent weaker tests from being mistaken for the theorem. The centre $(\beta, 1 + 2\beta, 1)$ gives the same type $[3, 8, 3]$ but only a group of order 2448, so fails generation. The centre $(1, 6, 1)$ has trace zero and admits an actual reversing semilinear map, so cannot pass the chirality test.

The exploratory normalization and sixteen completed intermediate cases are retained with their receipts. They are not an asserted census of all ranks or all types. In particular, our statement recovers the source's three reported positive pairs and does not turn the phrase "a computer search gave also three pairs" into an unproved assertion that there are no others. The positive claim is attributed to Leemans–Vandenschrick; the explicit centres and centralizer certificate are supplied here. No new question resolution, Lean proof, or numerical compression measurement is claimed.